

ADVANCED MANUFACTURING

10 Courses: 1 core mathematics course + 2 Group 1 courses + 3 courses from Group 1 or 2 + 4 technical electives

Group 1 Courses

535.628
Computer-Integrated Design & Manufacturing

Every Term

535.642
Control Systems for Mechanical Engineering Applications

Fall

535.659
Manufacturing System Analysis

Fall

535.660
Precision Mechanical Design

Fall/Spring

535.673
Mechanized Assembly: Hardware and Algorithms

Summer

535.674
Smart Manufacturing and Automation

Spring

535.721
Advanced Composite Materials & Manufacturing Processes

Spring/Summer

Group 2 Courses

535.737
Multiscale Modeling & Simulation of Mechanical Systems

Summer

535.684
Modern Polymeric Materials

Fall/Spring

535.720
Mechanics of Composite Materials and Structures

Fall

535.627
Computer-Aided Design

Every Term

535.744
AI for Mechanical Materials Design

Spring

Computational

Materials

535.618
Fabricatology: Advanced Materials Processing

Summer/Fall

535.638
Mechanical Packaging for Electronics Systems

Summer/Fall

535.733
Micromechanics of Heterogeneous Materials & Composites

Fall

535.672
Advanced Manufacturing Systems

Summer

515.658
Design for Additive Manufacturing

Fall

515.622
Micro and Nano Structured Materials & Devices

TBD

Applications